

SATYA BHARGAV MADDIPATI

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Education

University of Wisconsin–Madison

Madison, WI

M.S. in Computer Sciences — GPA: 4.0/4.0

Sep 2025 - May 2027

- **Coursework:** Software Engineering, Database Systems, Big Data, Computer Vision, Deep Learning, AI Agents

Indian Institute of Technology Bombay

Mumbai, India

B.Tech-M.Tech Dual Degree with Honors — GPA: 8.9/10

Aug 2018 - Jun 2023

Experience

Tata AIA Life Insurance | Software Engineer – Machine Learning

Mumbai, India

• [PYTHON](#) • [FASTAPI](#) • [REST APIS](#) • [MICROSERVICES](#) • [DOCKER](#) • [KUBERNETES](#) • [AZURE](#) • [CI/CD](#) Jun 2023 - Aug 2025

- Designed and deployed scalable **FastAPI** microservices backing a customer support platform serving **50K+** users on Azure, sustaining **0.8s p95 latency** and **99.5% uptime** under production load
- Owned design through delivery of production workflow services integrating **LangGraph/CrewAI** agents into existing claims systems, automating multi-step processing and cutting manual review effort by **35%**
- Built a **reusable, configurable training pipeline** that fine-tuned **5+ open-source LLMs (Llama 3.2, Qwen 2.5)** via **QLoRA/PEFT**, shipped with **automated tests and CI/CD** and validated through peer code review, reaching **94%** accuracy on internal benchmarks
- Designed a modular recommendation service over **2M+** customer records using embeddings and collaborative filtering, exposed as a **REST API** consumed by multiple internal teams

IIT Bombay | Graduate Research Assistant

Mumbai, India

• [PYTHON](#) • [DATA PIPELINES](#) • [PANDAS](#) • [NUMPY](#) • [TIME SERIES](#)

Apr 2022 - Jun 2023

- Built a modular forecasting pipeline for air-quality data: stochastic **KZ filtering** to strip meteorological trend, then swappable ensemble models (**XGBoost, Random Forests, SARIMA**) behind a common interface
- Ran **SHAP-based interpretability analysis** to quantify seasonal and feature-level contributions in the forecasts

Cognibot AI | Machine Learning Intern

Mumbai, India

• [COMPUTER VISION](#) • [OBJECT DETECTION](#) • [EFFICIENTDET](#) • [PYTHON](#)

Dec 2021 – Feb 2022

- Built a scalable people-tracking system for retail stores on an **EfficientDet** detector, profiling and tuning it across varied **compute and memory constraints** to meet real-time throughput targets

Projects

Reason-Guessr: Image Geolocation Web App | [HTML/CSS/JS](#) • [CLIP ViT-B/32](#)

reason-guessr.vercel.app

- Built and deployed an **interactive web application** that predicts GPS coordinates from a single street-level image, wiring a browser frontend to a Python inference service and shipping it to production on **Vercel**
- Backed by a **CLIP ViT-B/32** pipeline extracting interpretable cues (climate, plates, vegetation, architecture) zero-shot and fusing them with visual embeddings for coordinate regression; on a self-built **32K-image** dataset over **8,000** locations, cue fusion improved median haversine error from **90.6 km** to **88.9 km**

Search Distillation: Learning from a Teacher | [LLMS](#) • [SEARCH](#) • [DISTILLATION](#)

In progress

- Designing a search distillation study, with Dheeraj Dhillon, where the teacher has evidence access but the student must choose what to read rather than imitate the teacher's answers directly
- Comparing the student against plain distillation and information-gain search baselines on answer accuracy, retrieval cost, and abstention, starting with controlled synthetic tasks before moving to **MuSiQue** multi-hop questions

Grounded Vision-Language Agents for Instruction Following | [LLAVA-1.6](#) • [LORA/PEFT](#)

- Built a multimodal agent on **LLaVA-1.6 (7B)** with a custom **Observe→Reason→Act** loop for multi-step instruction following on **VisualWebBench** and **ScienceQA**, benchmarked against text-only **Mistral-7B + ReAct** baselines

Technical Skills

Programming: Java, JavaScript, HTML/CSS, Python, C++, SQL

Software & Web: REST APIs, FastAPI, Streamlit, Git, Unit/Integration Testing, Object-Oriented Design, PostgreSQL, MongoDB, Redis

Cloud & Infrastructure: Azure, AWS (EC2, S3, SageMaker), Docker, Kubernetes, CI/CD, Vercel

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face, OpenCV, LangChain, LangGraph, CrewAI, QLoRA, PEFT, Pandas, NumPy, SHAP, OpenAI/Anthropic/Gemini APIs

Achievements and Positions of Responsibility

Achievements: JEE Advanced Top 0.5% among 250K candidates; **Finalist**, Tata Innovation Workshop (**25+** teams) — WhatsApp AI cross-sell flow cutting add-on processing from **15 days to under 5 min**; Inter-IIT Sports Bronze Medal

Positions of Responsibility: Web Coordinator, Techfest IIT Bombay